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A Note to Students

Known by many as simply "Turabian" in honor of its original author, *A Manual for Writers of Research Papers, Theses, & Dissertations* ia the authoritative student resource on "Chicago style."

This book has helped generations of students successfully research, write, and submit papers in virtually all academic disciplines.

Its guidelines for source citations and style have been condensed and adapted for student writers from another, more comprehensive reference work, *The Chicago Manual of Style*.

- Part 1 covers every step of the research and writing process.

  It provides practical advice to help you formulate the right questions, read critically, and build arguments.

  It also shows you how to draft and revise your papers to strengthen both your arguments and your writing.

- Part 2 offers a comprehensive guide to the 2 methods of Chicago-style source citation, beginning with helpful information on general citation practices in Chap. 15.

  In the humanities and most social sciences, you will likely use the notes-bibliography style detailed in Chaps. 16–17; in the natural and physical sciences and some social sciences, you will more likely use the author-date style described in Chaps. 18–19.

- Part 3 covers Chicago's recommended editorial style, which will help you bring consistency to your writing in matters e.g. punctuation, capitalization, and abbreviations; this section also includes guidance on incorporating quotations into your writing and on properly presenting tables and figures.

- The appendix presents formatting and submission requirements for theses and dissertations that many academic institutions use as a model, but be sure to follow any local guidelines provided by your institution.

Preface

students writing research papers, theses, and dissertations in today's colleges and universities inhabit in world filled with digital technologies that were unimagined in 1937 - the year dissertation secretary Kate L. Turabian 1st assembled a booklet of guidelines for student writers at the University of Chicago.

The availability of word-processing software and new digital sources has changed the way students conduct research and write up the results.

But these technologies have not altered the basic task of the student writer: doing well-designed research and presenting it clearly and accurately while following accepted academic standards for citation, style, and format.

Turabian's 1937 booklet reflected guidelines found in an already classic resource for writers and editors published by the University of Chicago Press that would ultimately be known as *The Chicago Manual of Style* (*CMOS*).

The Press began distributing Turabian's booklet in 1947 and 1st published the work in book form in 1955, under the title *A Manual for Writers of Term Papers, Theses, and Dissertations*.

Over time, Turabian's book has become a standard reference for students of all levels at universities and colleges across the country.

Turabian died in 1987 at age 94, a few months after publication of the book's 5th edition.

Beginning with that edition, members of the Press editorial staff have carried out the revisions to the chapters on source citation, style, and paper format.

For the 7th edition (2007), Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams expanded the focus of the book by adding extensive new material adapted from their book *The Craft of Research*, also published by the University of Chicago Press and now in its 4th edition (2016).

Among the new topics covered in their chapters were the nature of research, finding and engaging sources, taking notes, developing an argument, drafting and revising, and presenting evidence in tables and figures.

Following the deaths of this remarkable trio of authors, whose collective voice will always animate this work, Joseph Bizup and William T. FitzGerald have with this edition assumed the mantle of revising their chapters for a new generation of students, as they had previously for *The Craft of Research*.

Part 1, now aligned with the most recent edition of that book, incorporates updated advice for writers and responds to recent developments in information literacy, including the use of digital materials.

Part 2 offers comprehensive guide to the 2 Chicago styles of source citation - the notes-bibliography format used widely in the humanities and most social sciences and the author-date format favored in many of the sciences and some social sciences.

Thoroughly updated guidance related to online citation practices have been supplemented throughout by new examples featuring the types of sources students are most likely to consult.

Part 3 addresses matters of spelling, punctuation, abbreviation, and treatment of numbers, names, special terms, and titles of works.

The final 2 chapters in this section treat the mechanics of using quotations and graphics (tables and figures), topics that are discussed from a rhetorical perspective in Part 1.

Both Parts 2 and 3 have been updated for this edition in accordance with the 17th edition (2017) of *The Chicago Manual of Style*.

The recommendations in this manual in some instances diverge from *CMOS* in small ways, to better suit the requirements of academic papers as opposed to published works.

The appendix presents guidelines for paper format and submission that have become the primary authority for dissertation offices throughout the United States.

These guidelines have been updated to reflect the nearly universal electronic submission of papers and to feature new examples from recently published dissertations.

This appendix is intended primarily for students writing PhD dissertations and master's and undergraduate theses, but the sections on format requirements and electronic file preparation will also aid those writing class papers.

An extensive bibliography, organized by subject area and fully updated, lists sources for research and style issues specific to various disciplines.

The guidelines in this manual offer practical solutions to a wide range of issues encountered by student writers, but they may be supplemented - or even overruled - by the conventions of specific disciplines or the preferences of particular

institutions, departments, or instructors.

All of the chapters on style and format remind students to review the requirements of their university, department, or instructor, which take precedence over the guidelines presented in this book.

Updating a book that has been used by millions of students over 80 years is no small task, and many people participated in preparing this 9th edition.

The University of Chicago Press Editorial Staff

Part I

Research & Writing - Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, Joseph Bizup, and William T. FitzGerald

Overview of Part I

The authors know how daunting it can feel to start a substantial research project, whether it's a doctoral dissertation, a master's or senior thesis, or just a long class paper.

But you can handle any project if you break it into its parts, then work on them 1 step at a time.

Part I of this book shows you how.

The authors 1st discuss the aims of research and what readers will expect of any research paper (a term the authors use broadly to refer to all varieties of *research-based writing*).

The authors then focus on how to find a research question and problem whose answer is worth your time and your readers' attention; how to find and use information from sources to back up your answer; and then how to plan, draft, and revise your paper so your readers will see that your answer is based on sound reasoning and reliable evidence.

Several themes run through this part.

- You can't plunge into a project blindly; you must plan it, then keep the whole process in mind as you take each step.

  So think big, but break the process down into small goals that you can meet one at a time.

- Your best research will begin with a question that *you* want to answer.

  But you must then imagine readers asking questions of their own: *So what if you don't answer it? Why should I care?*

- From the outset, you should try to write every day, not just to record the content of your sources but to clarify what you think of them.

  You should also write down your own developing ideas to get them out of the cozy warmth of your head and into the cold light of day, where you can see if they still make sense.

  You probably won't use much of this writing in your final draft, but it is essential preparation for it.

- No matter how carefully you do your research, readers will judge it by how well you present it, so you must know what they will look for in clearly written paper that earns their respect.

If you're an advanced researcher, skim Chaps. 1–4.

You will see there much that's familiar; but if you're also teaching, it may help you explain what you know to your students more effectively.

Many experienced researchers tell us that Chaps. 5–12 have helped them not only to explain to others how to conduct and report research, but also to draft and revise their own writing more quickly and effectively.

If you're just starting your career in research, you'll find every chapter of Part 1 useful.

Skim it all for an overview of the process; then as you work through your project, reread chapters relevant to your immediate task.

You may feel that the steps described here are too many to remember, but you can manage them if you take them one at a time, as you do do more research they'll become habits of mind.

Don't think, however, that you must follow these steps in exactly the other the authors present them.

Researchers regularly think ahead to future steps as they work through earlier ones and revisit earlier steps as they deal with a later one.

(That explains why we so often refer you ahead to anticipate a later stage in the process and back to revisit an earlier one.)

And even the most systematic researcher has unexpected insights that send her off in a new direction.

Work from a plan, but be ready to depart from it, even to discard it for a new one.

If you're a very new researcher, you may also think that some matters the authors discuss are beyond your immediate needs.

The authors know that a 10-page class paper differs from a dissertation.

But both require a kind of thinking that even the newest researcher can start practicing.

You begin your journey toward full competence when you not only know what lies ahead but also can start practicing the skills that experienced researchers began to learn when they were where you are now.

No book can prepare you for every aspect of every research project.

And this one won't help you with the specific methodologies in fields e.g. psychology, economics, and philosophy, much less physics, chemistry, and biology.

Nor does it tell you how to adapt what you learn about academic research to business or professional settings.

But it does provide an overview of the processes and habits of mind that underlie all research, wherever it's done, and of the plans you must make to assemble a paper, draft it, and revise it.

With that knowledge and with help from your teachers, you'll come to feel in control of your projects, not intimated by them, and eventually you'll learn to manage even the most complex projects on your own, in both the academic and the professional worlds.

The 1st step in learning the skills of sound research is to understand how experienced researchers think about its aims.

Chapter 1

What Research Is & How Researchers Think about It

Whenever we read about a scientific breakthrough or a crisis in world affairs, we benefit from the research of others, who likewise benefited from the research of countless others before them.

When we walk into a library, we are surrounded by more than 25 centuries of research.

When we go on the internet, we can read the work of millions of researchers who have posed questions beyond number, gathered untold amounts of information from the research of others to answer them, and then shared their answers with the rest of us.

We can carry on their work by asking and, we hope, answering new questions in turn.

Governments spend billions on research, businesses even more.

Research goes on in laboratories and libraries, in jungles and ocean depths, in caves and in outer space, in offices and, in the information age, even in our own homes.

Research is in fact the world's biggest industry.

So what, exactly, is it?

1.1 What Research Is

You already have a basis understanding of research: answering a question by obtaining information.

In this case, research can be as simple as choosing a new phone or as complex as discovering the origin of life.

In this book the authors use *research* in a specific way to mean a process of systematic inquiry to answer a question that not only the researcher but also others want to solve.

Research thus includes the steps involved in presenting or reporting it.

To be a true researcher, as we are using the term, you must share your findings and conclusions with others.

If you are new to research, you may think that your paper will add little to the world's knowledge.

But done well, it will add a lot to *your* knowledge and to your ability to communicate that knowledge.

As you learn to do your own research, you also learn to use and judge that of others.

In every profession, researchers must read and evaluate the work of others before they make a decision.

This is a job you will do better after you have learned how others judge yours.

This book focuses on research in the academic world, but every day we read or hear about research that affects our lives.

Often we get news of research secondhand, and it can be difficult to know what reasoning and evidence support a claim.

But research doesn't ask for our blind trust or that we accept something on the basis of authority.

It invites readers to think critically about evidence and reasoning.

That is how *research-based writing* differs from other kinds of persuasive writing: it must rest on shared facts that readers accept as truths independent of your feelings and beliefs.

Your readers must be able to follow your reasoning from evidence they accept to the claim you draw from it.

Your success as a researcher thus depends not just on how well you gather and analyze data but also on how clearly you report your reasoning so that your readers can test and judge it before making your claims part of their knowledge and understanding.

1.2 How Researchers Think about Their Aims

All researchers collect information, what we're calling *data*.

But researchers do not merely gather facts on a topic - *stories about the Battle of the Alamo*, e.g.

They look for specific data to test and support an answer to a question that their topic inspired them to ask, e.g. *Why has the Alamo story become a national legend?*

In doing so, they also imagine a community of readers who they believe will share their interest and help them test and support an answer to that question.

Experienced researchers, however, know that they must do more than convince us that their answer is sound.

They must also show us why their question was worth asking, how its answer helps us understand some bigger issue in a new way.

If we can figure out why the Alamo story has become a national legend, we might then answer a larger question: how have regional myths shaped the American character?

You can judge how closely your thinking tracks that of an experienced researcher by describing your project in a sentence like this:

1. **Topic**: I am working on X (*stories about the Battle of the Alamo*)
2. **Question**: because I want to find out Y (*why its story became a national legend*)
3. **Significance**: so that I can help others understand Z (*how such regional myths have shaped the American character*).

That sentence is worth a close look, because it describes not just the progress of your research but your personal growth as a researcher.

- **Topic**: "I am working on X ...": Those new to research often begin with a simple topic like *the Battle of the Alamo*.

  But too often they stop there, with nothing but a broad topic to guide their work.

  Beginning this way, they may pile up dozens or hundreds of notes but then can't decide what data to keep or discard.

  When it comes time to write, their papers become "data dumps" that leave readers wondering what all those data add up to.

- **Question**: "... because I want to find out Y ...": More experienced researchers begin not just with a topic but with a *research question*, e.g. *Why has the story of the Alamo become a national legend?*

  They know that readers will think their data add up to something only when they serve as evidence to support an answer.

  Indeed, only with a question can a researcher know what information to look for and, once obtained, what to keep - and not just data that support a particular answer but also data that test or discredit it.

  With sufficient evidence to support an answer, a researcher can respond to data that seem to contradict it.

  In writing a paper, the researcher tests that answer and invites others to test it too.

- **Significance**: "... so that I can help others understand Z": The best researchers understand that readers want to know not only that an answer is sound but also why the question is worth asking: *So what? Why should I care why the Alamo story has become a national legend?*

  Think of it this way: what will be lost if you *don't* answer your question?

  Your answer might be *Nothing. I just want to know.*

  Good enough to start but not to finish, because eventually your readers will want an answer beyond *Just curious.*

Answering *So what?* is tough for all researchers, beginning and experienced alike, because when you only have a question stemming from a topic of personal interest, it's hard to predict whether others will find its answer significant.

Some researchers therefore work backwards: they begin not by following their own curiosity but by crafting questions with implications for bigger ones that others in their field already care about.

But many researchers, including us, find that they cannot address that 3rd step until they finish a 1st draft.

So it's fine to *begin* your research without being able to answer *So what?*, and if you are a student, your teacher may even let you skip that last step.

But if you are doing advanced research, you *must* take it, because your answer to *So what?* is what makes your research matter to others.

In short, not all questions are equally good.

We might ask how many cats slept in the Alamo the night before the battle, but so what if we find out?

It is hard to see how an answer would help us think about any larger issue worth understanding, so it's a question that's probably not worth asking (though as we'll see, we could be wrong about that).

How good a question is depends on its significance to some community of readers.

Exactly what community depends on your field but also on how you frame your research.

You can try to expand your potential readership by connecting Z to even broader questions: *And if we can understand what has shaped the American character, we might understand better who Americans think they are. And when we know that, we might better understand why others in the world judge them as they do.*

Now perhaps political scientists will be as interested in this research as historians.

On the other hand, if you try to widen your audience too much, you risk losing it altogether.

Sometimes it's better to address a smaller community of specialists.

The authors can't tell you the right choice, but they can tell you 2 wrong ones: trying to interest everyone (some people just won't care no matter how you frame your research) or not trying to interest anyone at all.

1.3 Conversing with Your Readers

When you can explain the *significance* of your research, you enter into a kind of conversation with your research community.

Some people, when they think of research, imagine a lone scholar or scientist in a hushed library or lab.

But no places are more crowded with the presence of others than these.

When you read a book or an article or a report, you silently converse with its authors - and through them with everyone else th

In fact, every time you go to a written source for information, you join a conversation between writers and readers that began millennia ago.

And when you report your own research, you add your voice and hope that other voices will respond to you, so that you can in turn respond to them.

And so it goes.

Experienced researchers understand that they are participating in such conversations and that genuine research must matter not only to the researcher but also to others.

That is why the authors' formula - *I am working on X to find out Y so that others can better understand Z* - is so powerful: because it makes informing others the end of research.

But these silent conversations differ from the face-to-face conversations we have every day.

We can judge how well everyday conversations are going as we have them, and we can adjust our statements and behavior to repair mistakes and misunderstandings as they occur.

But in writing we don't have that opportunity: readers have to *imagine* writers in conversation with 1 another, as well as with themselves, and writers have to imagine their readers and their relationship to them.

In other words, writers have to offer readers a social contract: *I'll play my part if you play yours.*

Doing this is 1 of the toughest tasks for beginning researchers: get that relationship wrong and your readers will think you are naive or, worse, won't read your work at all.

Too many beginning researchers offer their readers a relationship that caricatures a bad classroom: *Teacher, I know less than you. So my role is to show you how many facts I can dig up. Yours is to say whether I've found enough to give me a good grade.*

Do that and you turn your project into a pointless drill, casting yourself in a role exactly opposite to that of a true researcher.

In true research, you must switch the roles of student and teacher.

You must imagine a relationship that goes beyond *Here are some facts I've dug up about 14th-century Tibetan weaving. Are they enough of the right ones?*

There are 3 better reasons to share what you've found.

You could say to your reader, *Here is some information that you may find interesting.*

This offer assumes, of course, that your reader wants to know.

You could also say not just *Here is something that should interest you* but *Here is something that will help you remedy a situation that troubles you.*

People do this kind of research every day in business, government, and the professions when they try to figure out how to address problems ranging from insomnia to falling profits to climate change.

In Chap. 2 the authors call such situations and their consequences *practical problems.*

When academic researchers address such practical problems, we say they are doing *applied* research.

Most commonly, though, academic researchers do *pure* research that addresses what we call *conceptual problems* - i.e., not troubling situations in the world but the limitations of our understanding of it (again see Chap. 2).

In this case, you say to your readers, *Here is something that will help you better understand something you care about.*

When you make this last sort of appeal, you imagine your readers as a community of receptive but also skeptical colleagues who are open to learning from you and even changing their minds - if you can make the case.

We now understand the *goal of research, at least in its pure form: it is not to have the last word but to keep the conversation going.*

The best question are those whose answers raise several more.
When that happens, everyone in the research community benefits.

Chapter 2

Defining a Project: *Topic, Question, Problem, Working Hypothesis*

A research project begins well before you search the internet or head for the library and continues long after you have collected all the data you think you need.

Every project involves countless specific tasks, so it is easy to get overwhelmed.

But in all research projects, you have just 5 general aims:

1. Ask a question worth answering.
2. Find an answer that you can support with good reasons.
3. Find good data that you can use as reliable evidence to support your reasons.
4. Draft an argument that makes a good case for your answer.
5. Revise that draft until readers will think you met the 1st 4 goals.

You might even post those 5 goals in your workspace.

Research projects would be much easier if we could march straight through these steps.

But you will discover (if you have not already) that the research process is not so straightforward.

Each task overlaps with others, and frequently you must go back to an earlier one.

The truth is, research is messy and unpredictable.

But that's also what makes it exciting and ultimately rewarding.

2.1 Find a Question in Your Topic

Researchers begin projects in different ways.

Many experienced researchers begin with a question that others in their field want to answer: *What caused the extinction of most large North American mammals?*

Others begin with just basic curiosity, a vague intellectual itch that they have to scratch.

They might not know what puzzles them about a topic, but they're willing to spend time to find out whether that topic can yield a question worth answering.

They realize, moreover, that the best research question is not one whose answer they want to know just for its own sake; it is one that helps them and others understand some larger issue.

E.g., if we knew why North American sloths disappeared, we might be able to answer a larger question that puzzles many historical anthropologists: *Did early Native Americans live in harmony with nature, as some believe, or did they hunt its largest creatures to extinction? And if we knew that, then we might also understand* ... (*So what?* again. See **1.2**)

Then there are those questions that just pop into a researcher's mind with no hint of where they'll lead, sometimes about matters so seemingly trivial that only the researcher thinks they're worth answering: *Why does a coffee spill dry up in the form of a ring?*

Such a question might lead nowhere, but you can't know that until you see its answer.

In fact, the scientist puzzled by coffee rings made discoveries about the behavior of fluids that others in this field thought important - and that paint manufacturers found valuable.

If you cultivate the ability to see what's odd in the commonplace, you'll never lack for research projects as either a student or a professional.

If you already have a focused topic, you might skip to **2.1.3** and begin asking questions about it.
If you already have some questions, skip to **2.1.4** to test them using the criteria listed there.
Otherwise, here's a plan to help you search for a topic.

2.1.1 Search Your Interests

Beginning researchers often find it hard to pick a topic or believe they lack the expertise to research a topic they have.

But a research topic is an interest stated specifically enough for you to imagine *becoming* a local expert on it.

That doesn't mean you *already* know a lot about it or that you'll know more about it than others, including a teacher or advisor.

You just want to know more about it than you do now.

If you can work on any topic, the authors offer only a cliché: start with what interests you.
Ask these questions:

- What special interests do you have - chess, old comic books, scouting?

  The less common, the better.

  Choose one and investigate something about it that you don't know.

- Where would you like to travel?

  Find out all you can about your destination.

  What particular aspect surprises you or makes you want to learn more?

- Can you find an online discussion list or social media page focused on issues that interest you?

- Visit a museum or a "virtual museum" on the internet with exhibitions that appeal to you.

  What catches your interest that you would like to know more about?

- Have you taken positions on issues in your fields or in debates with others but found that you couldn't back up your views with good reasons and evidence?

- What issues in your field do people outside your field misunderstand?

- What topic is your instructor or advisor working on?

  Would she like you to explore a part of it?

  Don't be too shy to ask.

- Does your library have rich resources in some field?

  Ask your instructor or a librarian.

- What intrigues you in your reading?

  What connections do you see among different things you are reading?

- What other courses will you take in your field or out of it?

  Find a textbook and skim it for study questions.

- If you have a job in mind, what kind of writing might help you get it?

  Employers often ask for samples of an applicant's work.

Once you have a list of possible topics, choose 1 or 2 that interest you most and explore their research potential.

Sometimes beginning researchers choose a topic because they already know what they want to say about it, even before they've done any research.

That's a mistake: the best topics provoke good questions; the worst come with ready-made answers.

To gauge a topic's potential, do these things:

- In the library, look up your topic in a general guide e.g. *CQ Researcher* and skim the subheadings.

  In an online database e.g. Academic Search Premier, you can explore your topic through subject terms.

  If you have a narrower focus, you can do the same with specialized guides e.g. *Women's Studies International*.

  At most libraries today, such guides are found online.

- On the internet, google your topic, but don't surf indiscriminately.

  Look 1st for websites that are roughly like the sources you would find in a library, e.g. online encyclopedias.

  Read the entries on your general topic, an then copy their lists of references for a closer look.

  Few experienced researchers trust Wikipedia as a reliable source to cite as evidence, but most would *use the site to find ideas and more specific sources.*

- Finally, think ahead: you may be in for a long relationship with your topic, so be sure it interests you enough to get you through the inevitable rocky stretches.

2.1.2 Make Your Topic Manageable

If you pick a topic that sounds like an encyclopedia entry - *bridges, birds, masks* - you'll find so many sources that you could spend a lifetime reading them.

You must carve out of your topic a manageable piece.

Before you start researching, limit your topic to reflect a special interest in it: *What is it about, say, masks that made you choose them? What particular aspect of them interests or puzzles you?*

Think about your topic in a context that you know something about, and then add words and phrases to reflect that knowledge:

Example 2.1.1. *masks in religious ceremonies*
masks as symbols in Hopi religious ceremonies
mudhead masks as symbols of sky spirits in Hopi fertility ceremonies

You might not be able to focus your topic until after you start reading about it.

That takes time, so start early (you can do much of this preliminary work online):

- Begin with an overview of your topic in a general encyclopedia (in the bibliography, see items in category 2 in the general sources); then read about it in a specialized one (see items in category 2 in your field).

- Skim a survey of your topic (encyclopedia entries usually cite a few).

- Skim subheads under your topic in an annual bibliography in your field (in the bibliography, see items in category 4 in your field).

  That will also give you a start on a reading list.

- Search the internet for the topic (but evaluate the reliability of what you find; see **3.3.2**).

Especially useful are topics that spark debate: *Fisher claims that Halloween masks reveal children's archetypal fears, but do they?*

Even if you can't resolve the debate, you can learn how such debates are conducted (for more on this, see **3.1.2**).

2.1.3 Question Your Topic

Once they have a focused topic, many new researchers start plowing through all the sources they can find, taking notes on everything they read.

Then they dump it all into a report with little sense of purpose or direction.

Experienced researchers, however, document information not for its own sake but to support an answer to a question they (and they hope their readers) think worth asking.

So the best way to begin working on a focused topic is to pose questions that direct you to just the information you need to answer them.

Do this not just once, early on, but throughout your project.

Ask questions as you read, especially *how* and *why* (see also **4.1.1**–**4.1.2**).

Try the following kinds of questions (the categories are loose and overlap, so don't worry about keeping them distinct).

1. Ask how the topic fits into a larger context (historical, social, cultural, geographic, functional, economic, and so on):

   - How does your topic fit into a larger story?

Example 2.1.2. *What cam before masks?*
How did masks come into being?
Why?
What changes have they caused in other parts of their social or geographic setting?
How and why did that happen?
Why have masks become a part of Halloween?
How and why have masks helped make Halloween the biggest American holiday after Christmas?

- How is your topic a functioning part of a larger system?

  Example 2.1.3. *How do masks reflect the values of specific societies and cultures?*
  What roles do masks play in Hopi dances?
  In scary movies?
  In masquerade parties?
  For what purposes are masks used other than disguise?
  How has the booming market for kachina masks influenced traditional designs?

- How does your topic compare to and contrast with other topics like it?

  Example 2.1.4. *How do masks in Native American ceremonies differ from those in Africa?*
  What do Halloween masks have to do with Mardi Gras masks?
  How are masks and cosmetic surgery alike?

2. Ask questions about the nature of the thing itself, as an independent entity:

   - How has your topic changed through time? Why? What is its future?

     Example 2.1.5. *How have Halloween masks changed?*
     Why?
     How have Native American masks changed?
     Why?

   - How do the parts of your topic fit together as a system?

     Example 2.1.6. *What parts of a mask are most significant in Hopi ceremonies?*
     Why?
     Why do some masks cover only the eyes?
     Why do so few masks cover just the bottom half of the face?

   - How many different categories of your topic are there?

     Example 2.1.7. *What are the different kinds of Halloween masks?*
     What are the different qualities of masks?
     What are the different functions of Halloween masks?

3. Turn positive questions into a negative ones:

   Example 2.1.8. *Why have masks* not *become a part of Christmas?*
   How do Native American masks not *differ from those in Africa?*
   What parts of masks are typically not *significant in religious ceremonies?*

4. Ask speculative questions:

   Example 2.1.9. *Why are masks common in African religions but not in Western ones?*
   Why are children more comfortable wearing Halloween masks than are most adults?
   Why don't hunters in camouflage wear masks?

5. Ask *What is?* questions: how would things be different if your topic never existed, disappeared, or were put into a new context?

Example 2.1.10. *What if no one ever wore masks except for safety reasons?*

What if everyone wore masks in public?

What if movies and TV were like Greek plays and all the actors wore masks?

What if it were customary to wear masks on blind dates?

In marriage ceremonies?

At funerals?

6. Ask questions that reflect disagreements with a source: if a source makes a claim you think is only weakly supported or even wrong, make that disagreement a question (see also **4.1.2.**).

   Example 2.1.11. *Martinez claims that carnival masks uniquely allow wearers to escape social norms.*

   But I think religious masks also allow wearers to escape from the material realm to the spiritual.

   Is there a larger pattern of all masks creating a sense of alternative forms of social or spiritual life?

7. Ask question that build on agreement: if a source offers a claim you think is persuasive, ask questions that extend its reach (see also **4.1.1**).

   Example 2.1.12. *Elias shows that masked balls became popular in 18th-century London in response to anxiety about social mobility.*

   Is the same anxiety responsible for similar developments in other European capitals?

   You can also ask a question that supports the same claim with additional evidence.

   Example 2.1.13. *Elias supports his claim about masked balls entirely with published sources.*

   It is also supported by evidence from unpublished sources e.g. letters and diaries?

8. Ask questions analogous to those that others have asked about similar topics.

   Example 2.1.14. *Smith analyzed the Battle of Gettysburg from an economic point of view.*

   What would an economic analysis of the Battle of the Alamo turn up?

9. Look for questions that other researchers pose but don't answer.

   Many journal articles end with a paragraph or 2 about open questions, ideas for more research, and so on.

   You might not be able to do all the research they suggest, but you might carve out a piece of it.

10. Find a professional discussion forum on your topic, then "lurk," just reading the exchanges to understand the kinds of questions being asked.

    If you can;t find one using a search engine, ask a teacher or visit websites of professional organizations in your field.

    Look for questions that spark your interest.

    If questions from students are welcomed, you can even post one yourself, so long as it is very specific and narrowly focused.

2.1.4 Evaluate Your Questions

After asking all the questions you can think of, evaluate them.

Not all questions are equally good.
Look for questions whose answers might make you (and your readers) think about your topic in a new way.
Avoid questions like these:

- Their answers are settled fact that you could just look up.

  Example 2.1.15. *What was Audre Lorde's 1st published poem?*

  Question that ask *how* and *why* call for interpretations, not just the discovery of facts.

  That's why they invite deeper thinking than questions beginning *who, what, when*, or *where*, and deeper thinking leads to more interesting answers.

- Their answers can't be plausibly disproved.

  Example 2.1.16. *How important are masks in Inuit culture?*

  The answer is obvious: *Very.*

  If you can't imagine disproving a claim, then proving it is pointless.

  (On the other hand, world-class reputations have been won by those who questions a claim that seemed self-evidently true - e.g., that the sun circled the earth - and dared to disprove it.)

- Their answers would be merely speculative.

  Example 2.1.17. *Would church services be as well attended if the congregation all wore mask?*

  If you can't imagine finding data that would settle the question, it's not a question you can answer.

- Their answers are dead ends.

  Example 2.1.18. *How many black cats slept in the Alamo the night before the battle?*

  It's hard to see how an answer would help us think about any larger issue worth understanding better, so the question is probably not worth asking.

- Their answers require different capacities from the ones you have.

  Example 2.1.19. *How do Japanese translations of* The Great Gatsby *treat early 20th-century America?*

  If you can't read Japanese, this question is not for you to answer.

- The answers require more or different resources - materials, technology, money, especially time - than you have.

  Example 2.1.20. *How is childhood represented in the Victorian novel?*

  Can you read enough of them in the time you have to arrive at a reasonable answer?

Don't reject a question because you think someone must already have asked it.

Until you know, pursue its answer as if you asked 1st.

Even if someone has answered it, you might come up with a better answer or at least one with a new slant.

In fact, in the humanities and social sciences the best questions usually have more than 1 good answer.

You can also organize your project around comparing and contrasting competing answers and supporting the best one (see **6.2.5**).

The point is to find a question that *you* want to answer.

Too many students, both graduate and undergraduate, think that the aim of education is to learn settled answers to someone else's questions.

It's not.

It is to find your own answers to your own questions.

To do that, you must learn to wonder about things, to let them puzzle you - particularly things that seem commonplace.

2.2 Understanding Research Problems

In Chap. 1 the authors gave you a formula that expresses how experienced researchers think about their work:

1. **Topic**: I am working on X (*stories about the Battle of the Alamo*)
2. **Question**: because I want to find out Y (*why its story became a national legend*)
3. **Significance**: so that I can help others understand Z (*how such regional myths have shaped the American character*).

When you can state that significance from the point of view of your readers, you have more than a question: you have posed a *research problem* that they recognize needs a solution.

Among researchers, the term *problem* has a special meaning that sometimes confuses beginners.

In our everyday world, a problem is something we try to avoid.

But in academic research, a problem is something we seek out, even invent.

Indeed, without a problem to work on, a research is out of work.

Experienced researchers often talk about their problems in shorthand.

When asked what they are working on, they often answer with what sounds like a general topic: *adult measles, mating calls of Wyoming elk.*

As a result, beginners may think that having a topic to read about is the same thing as having a problem to solve.

But without a specific question to answer and a reason to find that answer significant, researchers have no way of knowing when they have enough.

So they can be tempted to throw in everything just to be safe.

To avoid the judgment that your paper is just a *data dump*, you need a problem, one that focuses on finding just those data that will help you solve it.

To find one, you need to know how problems work.

2.2.1 Understanding Practical and Conceptual Problems

There are 2 kinds of research problems: *practical* and *conceptual*.

Each of them has a 2-part structure:

- a situation or *condition*, and
- undesirable *costs* or *consequences* caused by that condition.

Your research question is about your problem's condition; its significance follows from your problem's cost or consequence.

What differentiates practical and conceptual problems is the nature of those conditions and costs/consequences.

The condition of a practical problem can be any state of affairs in the world that troubles you or, better, your readers: a traffic jam, foreign competition, a disease we can't effectively treat.

The cost of a practical problem is always some tangible effect we don't like: inconvenience, expense, pain, even death.

Practical problems are often a matter of perspective: if my company's products are outselling yours, that's problem for you but not for me.

The condition of a conceptual problem is always some version of not knowing or understanding something.

A conceptual problem does not have a tangible cost but a consequence.

This consequence is a particular kind of ignorance: a lack of understanding that keeps us from understanding something else that is even more significant.

Put another way, because we haven't answered 1 question, we can't answer another that is more important.

In short, practical problems concern what we should *do*; conceptual problems concern what we should *think*.

Practical problems are most common in the professional world; conceptual problems are most common in academe.

2.2.2 Distinguishing Pure & Applied Research

We call research *pure* when it addresses a conceptual problem that does not have any direct practical consequences, when it only improves the understanding of a community of researchers.

We call research *applied* when it addresses a conceptual problem that does have practical consequences.

You can tell whether research is pure or applied by considering the significance of your project: *is it about understanding or doing?*

1. **Topic**: I am studying how readings from the Hubble telescope differ from readings for the same stars measured by earthbound telescopes
2. **Question**: because I want to find out how much the atmosphere distorts measurements of electromagnetic radiation
3. **Practical Significance**: so that astronomers can use data from earthbound telescopes to *measure* more accurately the density of electromagnetic radiation.

Applied research is common in academic fields e.g. business, engineering, and medicine and in companies and government agencies that do research to understand what must be known before they can solve a problem.

Some new researchers may be uneasy with pure research because the consequence of a conceptual problem - not knowing something - seems so abstract.

Since they are not yet part of a community that cares deeply about understanding its part of the world, they feel that their findings aren't good for much.

So they try to show the importance of their conceptual answer by cobbling onto it an implausible practical use:

1. **Topic**: I am studying differences among 19th-century versions of the Alamo story
2. **Question**: because I want to find out how politicians used stories of such events to shape public opinion
3. **Potential Practical Significance**: in order to protect ourselves from unscrupulous politicians.

Most readers will find the link between this research question and its asserted significance a stretch.

But for researchers in American history, the question does not need to have practical significance.

As the term *pure* suggests, many researchers value the pursuit of knowledge "for its own sake" as a reflection of humanity's highest calling to know more.

So if you are doing academic research, resist the urge to turn a conceptual problem into a practical one - unless you've specifically been asked to do so.

You are unlikely to solve any genuine practical problem in a course project.

And in any case, most academic researchers see their mission not as fixing the problems of the world but as understanding them better (which may or may not lead to fixing them).

2.3 Propose a Working Hypothesis

Before you get deep into your project, try 1 more step.

It is one that some beginners resist but that experienced researchers usually attempt.

Once you have a question, imagine some plausible answers, no matter how sketchy or speculative.

At this stage, don't worry whether they're right.

That comes later.

E.g., suppose you ask, *Why do some religions use masks in ceremonies while others don't?*

You might speculate:

Example 2.3.1. *Maybe cultures with many spirits need masks to distinguish them.*

Maybe masks are common in cultures that mix religion and medicine.

Maybe religions originating the Middle East were influenced by the Jewish prohibition against idolatry.

Even a general answer can suggest something worth studying:

Example 2.3.2. *Maybe it has to do with the role of masks in nonreligious areas of a culture.*

Try to imagine at least 1 plausible answer, no matter how tentative or speculative.

If 1 answer seems promising, call it your *working hypothesis* and use it to guide your research.

You can, of course, start gathering data with no more than a question to guide you, but even a tentative working hypothesis will help you think about the kind of data you'll need as evidence to support it: numbers? quotations? observations? images? historical facts?

In fact, until you have a hypothesis, you can't know whether any data you collect are relevant to your project.

If you can't imagine any working hypotheses, reconsider your question.

You may even decide to start over with a new topic.

That costs time in the short run, but it may save you from a failed project.

Under no circumstances should you put off thinking about a hypothesis until you begin drafting your paper or, worse, until you've almost finished it.

You might not settle on the *best* answer to your question until you're well into writing your paper, for writing is an act of discovery.

But you can't wait until that last draft to start thinking about *some* answer.

2.3.1 Beware the Risks in a Working Hypothesis

Don't settle on a final answer too soon: working hypotheses are meant to change.

But many new researchers and some experienced ones are afraid to consider *nay* working hypothesis early in their project, even one they hold lightly, because they fear it might bias their thinking.

There is some risk of that, but a working hypothesis need not close your mind to a better one.

Even the most objective scientist devises an experience to test for just a few predicted outcomes, often just one.

A working hypothesis is a risk only if it blinds you to a better one or if you can't give it up when the evidence says you should.

So as in all relationships, don't fall too hard for your 1st hypothesis: the more you like it, the less easily you'll see its flaws.

Despite that risk, it's better to start with a flawed hypothesis than with none at all.

2.3.2 If You Can't Find an Answer, Argue for Your Question

We have focused so much on questions that you might think your project fails if you can't answer yours.

Not so.

Much important research explains why a question no one has yet asked should be asked: *Do turtles dream? Why is yawning contagious?*

Papers addressing such questions don't argue for answers; they explain why the question is important and what a good answer might look like.

Or perhaps you find that someone has answered your question, but incompletely or even - if you're lucky - incorrectly.

If you can't find the right answer, you still help your research community by showing that a widely accepted one is wrong.

You can even organize your paper around a working hypothesis you abandon.

If after lots of research, you can't confirm it, you can explain why that answer seemed reasonable at the time but turned out to be wrong and so isn't worth the time of other researchers.

That in itself can be a valuable contribution to the conversation on your topic.

(See 10.1.1–10.1.2 for how to use an apparently good idea that turns out to be wrong.)

Only by asking question after question will you develop the critical imagination you need to excel at research.

Experienced researchers know there are few, if any, final answers, because there are no final questions.

They know that it's as important to ask a new question as it is to answer an old one, and that 1 day their new question will become old and yield to a newer researcher's still newer one.

That's how the conversations of research communities progress.

2.4 Build a Storyboard to Plan & Guide Your Work

For a short paper, you might not need a detailed plan - a sketch of an outline might do.

But for a long project, you'll usually need more, especially for one as long as a thesis or dissertation.

The 1st plan that comes to mind is usually an outline, with its *I*s and *II*s and *A*s and *B*s and so on (See **23.4.2.2**).

If you prefer an outline, use one, especially if your project is relatively short.

The problem is that an outline can force you to specify too much too soon and so lock up a final form before you've done your best thinking.

To avoid that risk, any researchers, including those outside the academic world, use a *storyboard* to plan and guide their work.

A storyboard is like an outline spread over several pages, with lots of spae for adding data and ideas as you go.

But it is more flexible than an outline because storyboard pages can be moved around, allowing you to try out new ways of organizing your ideas.

You can spread its pages across a wall, group related pages, and put minor sections below major ones to create a "picture" of your project that shows you at a glance the design of the whole and your progress through it.

For this reason, a storyboard is useful at every stage of your project.

It can help you plan your research, develop your argument, organize your paper, write a 1st draft, and test a final one.

Someday you may have the time to amble through sources, reading just what interests you.

Such random browsing has opened up many important lines of research.

But if your paper is due in a month or so, or even sooner, you need a plan.

A storyboard is a simple and reliable device to help you create one.

2.4.1 State Your Question & Working Hypotheses

To start a storyboard, state at the top of its 1st page your question and working hypothesis as precisely as you can.

Then add plausible alternatives and new hypotheses as you think of them, and cross off those you prove wrong.

But save them, because you might be able to use 1 of them in your introduction (see **10.1.1**).

2.4.2 State Your Reasons

The authors say more about the structure of arguments in Chap. 5.

For now, the commonsense understanding of an argument as a claim supported by reasons and evidence is enough.

Put at the top of separate pages each reason that might support your best hypothesis, even if you have only 1 or 2.

Imagine explaining your project to a friend.

You say, *I want to show that Alamo stories helped develop a unique Texan identity*, and your friend asks, *Why do you think so?*

Your reasons are the general statements that you offer to support your answer: *Well, 1st, the stories distorted facts to emphasize what became central to Texan identity; 2nd, the stories were 1st used to show that Texas (and the Wild West) was a new kind of frontier; 3rd, . . .* and so on.

If you can't think of more than 1 or 2 reasons, put placeholders at the tops of pages: *Reason 3: Something about Alamo stories making Texans feel special.*

If you know only *how* you want a reason to support your answer, state that: *Reason 4: Something to show that Alamo stories were more than just myth.*

Each reason, of course, needs support, so for each reason, ask, *Why do I think that? What evidence will I need to prove it?*

That will help you focus your search for evidence (see **2.4.3** and **5.4.2**).

If you're new to your topic or early in your project, your reasons may be only educated guesses that you'll later change.

But a list of reasons, no matter how speculative, cannot only guide your research but also focus your thinking and help you anticipate the argument you will eventually make.

2.4.3 Sketch the Kind of Evidence You Should Look For

Every field prefers its own kinds of evidence - numbers, quotations, observations, historical facts, images, and so on.

So for each reason, sketch the kind of evidence that you think you'll need to support it.

Even imagine what the most convincing evidence would look like.

If you can't imagine the kind of evidence you'll need, leave that part of the page blank, then read secondary sources to find out the kind of evidence researchers in your field typically use (see **3.1.2**).

2.4.4 Look at the Whole

Lay the pages on a table or tape them on a wall.

Then step back and look at their order.

When you plan a 1st draft, you must put its parts in some order, so you might as well think about one now.

Can you see a logic in your storyboard?

Cause and effect?

Narrative time?

Relative importance?

Complexity?

Length?

(See **6.2.5** for more principles of order.)

Try out different orders.

This storyboard isn't your final plan; it's only a tool to guide your thinking and organize what you find.

When you fill a page, try drafting that section, because writing out your ideas can improve your thinking at every stage of your project.

2.5 Join or Organize a Writing Group

1 of the best ways to stay on track with your project is to join or organize a writing group.

In many fields, especially in the humanities and social sciences, scholars read, think, and write mostly alone.

But it doesn't have to be that way, at least not entirely.

Find someone other than your instructor or advisor to talk with about your progress, to review your drafts, even to pester you about how much you have written.

That person might be a generous friend or, better, another writer with whom you can trade feedback on ideas and drafts.

Better yet is a writing group: 4 or 5 people working on their own projects who meet regularly to discuss each other's work.

Early on, start each meeting with a summary of each person's project in that 3-part sentence: *I'm working on the topic X, because I want to find out Y, so that I (and you) can better understand Z.*

As your projects develop, start with an "elevator story," a short summary of your research that you might give someone in the elevator on the way to the meeting.

It should include that 3-part sentence, a working hypothesis, and the major reasons supporting it (see **13.4**).

In later stages, share outlines and drafts so that the members of the group can serve as surrogate readers.

If your group has a problem with your draft, so will your final readers.

Your group can even help you brainstorm when you bog down.

All of this support is valuable.

But for many writers, a writing group is valuable simply for the discipline it imposes.

It is easier to meet a schedule when you know you must report your progress to others.

Writing groups are standard practice for those preparing theses or dissertations.

But the rules may differ for class papers.

Some teachers worry that writing groups or writing partners might provide more help than they should.

So if you join a group, be clear with your teacher about what it will do.

If you don't she may decide the assistance you have received is inappropriate (see **7.10**).

Chapter 3

Finding Useful Sources

Once you have a research question and perhaps a working hypothesis, you can start looking for the data you'll need to test your hypothesis and develop your argument.

In this chapter the authors explain how to find those data in sources and in the next how to work with them.

But don't think of finding sources and reading them as separate steps.

Once you have a promising source, read it to find other sources.

And as you fill your storyboard with notes, you'll discover gaps and new questions that only more sources can fill.

So while we discuss finding and using sources as distinct steps, you'll more often do them repeatedly and simultaneously.

3.1 3 Kinds of Sources & Their Uses

Sources are conventionally categorized into 3 kinds: *primary, secondary*, and *tertiary* (think 1st-, 2nd-, and 3rdhand).

Their boundaries are fuzzy, but knowing these categories can help you conduct your research.

3.1.1 Consult Primary Sources for Evidence

Primary sources are "original" materials that provide you with the "raw" data or evidence you will use to develop, test, and ultimately justify your hypothesis or claim.

In history, primary sources are artifacts or documents that come directly from the period you are studying: letters, diaries, objects, maps, even clothing.

In literature or philosophy, your main primary sources are usually the texts you are analyzing, and your data are the words on the page.

In the arts, your primary source would be the works of art you are interpreting.

In social sciences, e.g. sociology or political science, census or survey data would also count as primary sources, as would data obtained through observation or experiment in many fields.

The primary sources for those collected data are reports of original research in scholarly journals or government and commercial databases.

3.1.2 Read Secondary Sources to Learn from Other Researchers

Secondary sources are books, articles, papers, or reports that are based on primary sources and intended for scholarly or professional audiences.

An article in a scholarly journal analyzing Alamo stories would be a secondary source for researchers working on those stories.

The body of secondary sources in a field is sometimes called that field's *literature.*

The best secondary sources are books from reputable university presses and articles or reports that have been *peer reviewed*, meaning that they were vetted by experts in the field before they were published (see **3.3.2**).

Secondary sources also include specialized encyclopedias and dictionaries that offer essays written by scholars in a field.

You use secondary sources for 3 purposes:

1. *To keep up with current research.*

   Researchers read secondary sources to keep up with the work of other researchers, to inform and refine their thinking, and to motivate their own work by adding to a published line of research.

2. *To find other points of view.*

   A research paper is not complete until the researcher acknowledges and responds to the views of others and to this readers' predictable questions and disagreements.

   You can find most of those other points of view in secondary sources.

   What alternatives to your ideas do they offer?

   What evidence do they cite that you must acknowledge?

   Some new researchers think they weaken their case if they mention any view opposing their own.

   The opposite is the truth.

   When you acknowledge competing views, you show readers that you not only know those views but can confidently respond to them.

   (For more on this, see **5.4.3**.)

3. *To find models for your own research and analysis.*

   You can use secondary sources to find out not just *what* others have written about your topic but *how* they have written about it, as models for the form and style of your own paper.

   Imagine a secondary source as a colleague talking to you about your topic.

   As your respond, you'd want to sound like someone who knows the field, and so you'd try to learn how she reasons, the language she uses, the kinds of evidence she offers, and the kinds she rarely or never uses.

   The "conversation" would be in writing, so you'd even imitate stylistic details e.g. whether she writes in long paragraphs or breaks up her pages with subheads and bullet points (common in the social sciences, less common in the humanities).

You can also use a secondary source as a model for your conceptual analysis.

If, e.g., you were analyzing Alamo stories, you might study how a source treats Custer's Last Stand.

Is its approach psychological, social, historical, political?

Its particular reasons or evidence will probably be irrelevant to your project, but you might support your answer with the same kinds of data and reasoning, perhaps even following the same organization.

So if you come across a source that's not exactly on your topic but treats one like it, skim it to see how that researcher thinks about his material and presents it.

(You don't have to cite that source if you use only its general logic, but you may cite it to give your own approach more authority.)

You can even borrow evidence from secondary sources to use in your own arguments, but you should do so only if you do not have access to the primary sources from which that evidence was originally taken.

If you're doing advanced work, check the accuracy of important quotations, facts, or numbers you draw from secondary sources.

Of course, if you were studying how the Alamo story has been analyzed, then secondary sources offering those analyzes would be your primary sources.

If you're new to a field, you may find secondary sources hard to read: they assume a lot of background knowledge, and many aren't clearly written (see **11.2**).

If you're working on a topic new to you, you might begin with an overview in a specialized encyclopedia or reliable tertiary source.

3.1.3 Read Tertiary Sources for Introductory Overviews

Tertiary sources are books and articles that synthesize secondary sources for general readers.

They include textbooks, encyclopedias (including Wikipedia), and dictionaries, as well as articles in publications for broad audiences, like *Time* and the *Atlantic*.

In the early stages of research, you can use tertiary sources to get a broad overview of your topic.

But if you are making a scholarly argument, you should rely on secondary sources because these make up the conversation in which you are seeking to participate.

If you cite tertiary sources in a scholarly argument, you will mark yourself as a novice or outsider, and many readers won't take you - or your argument - seriously.

3.2 Search for Sources Systematically

Knowing where to begin your search for sources can be overwhelming at 1st.

It is temping to simply enter terms into a familiar search engine (e.g., Google) and see what comes up.

The authors do this too, but they also know that there are more systematic and productive ways for discovering useful, credible sources.

Make the library the focus of your search strategies even as you draw on the expertise of others inside or outside the academic community and use the resources of the internet.

3.2.1 Ask a Librarian

The best advice the authors can offer is to draw on the research expertise of librarians.

Both general reference librarians and (in larger libraries) subject area specialists can help you refine your search parameters and direct you to the right tools for your specific research question.

They can help you use the catalog to locate materials held by your library or by other libraries (and obtainable through interlibrary loan).

These same librarians typically design research guides that identify reference works and online databases for specific fields.

If you're a new researcher, seize every opportunity to learn online search techniques in your field.

And don't by shy.

Librarians love to assist researchers of all levels and at all stages of the research process.

They can help you formulate your research question and plan, develop search terms, and inventory your results to ensure you haven't overlooked something of value.

The only embarrassing question is the one you failed to ask.

If you already have a research question, share it: *I'm looking for data on X because I want to find out . . .*

If you have a working hypothesis and reasons, share them too: *I'm looking for data to show Y [your reason] because I want to claim Z [your hypothesis].*

Rehearse your questions to avoid wasting your time and theirs.

3.2.2 Consult Reference Works

Researchers in all fields share common values and habits of thought, but every field has its own ways of doing things.

To learn about the ways of your field, browse the shelves in your library's reference room that hold guides to your field's particular research methods, databases, and special resources (in the bibliography, see items in category 3 in your field).

At least familiarize yourself with the following resources (in the bibliography, see category 4 for lists of sources in your field; many are also online):

- a bibliography of works published each year in your field, e.g. *Philosopher's Index* or *Education Index*
- summary bibliographies of works on a specific topic collected over several years (*Bibliographic Index* is a bibliography of bibliographies)
- annual literature reviews; look for a title in your field beginning with *Reviews in . . .*
- specialized reference works e.g. the *Encyclopedia of Philosophy* or the *Concise Oxford Dictionary of Literary Terms*, where you may find an overview of your topic and often a list of standard primary and secondary sources (in the bibliography, see items in categories 1–2 in your field).

3.2.3 Explore Online Databases

What sets libraries apart from the internet are subscriptions to indexes and databases.

After books, these are arguably a library's most valuable assets, since they give researchers access to materials they could not obtain otherwise.

Each library's subscription will differ, with major research libraries offering the most comprehensive access to specialized indexes and databases.

However, academic libraries and many public libraries offer powerful online tools that greatly extend their actual collections.

You will certainly want to make use of these general and specialized resources in your research.

At least become familiar with the major databases to which your library subscribes, e.g. Academic Search Premier, MLA International Bibliography, or PubMed.

Many academic databases either provide abstracts or direct you to articles that include abstracts.

Looking at these can help you decide if an article itself is worth reading carefully.

Some databases allow you to access full-text articles or books.

But beware: if you library does not subscribe to a particular journal included in a database, you might be asked to pay a fee to access a full-text article.

Before doing so, *always* speak with a librarian about other means of access.

3.2.4 Search Your Library Catalog

In your research, you will likely use your library's catalog in 2 ways: keyword searching and browsing.

When you have examined some sources to identify a list of *keywords* related to your topic, you are ready to use these terms to search the catalog.

In most libraries you must choose the category (books, articles, journals, etc.) you wish to use for your search.

If your sources include books, you can use the Library of Congress subject headings, found either on the back of their little page or on their "details" page in the online catalog, to search for related materials.

E.g., the online entry for this book includes these 2 topics:

Example 3.2.1. *1. Dissertations, Academic. 2. Academic writing.*

If you search an online catalog, you will find all the books on that subject in that library.

A book may be cross-listed under multiple subject headings.

In that case, look at titles listed under those headings as well.

You can also *browse* the catalog for books with similar *call numbers*.

Also search your online catalog using keywords from your question or working hypothesis - *Alamo, Texas independence, James Bowie*.

If you find too many titles, start with those published in the last 10 or 15 years by well-known university presses.

For a wide selection, search WorldCat if your library subscribes.

Otherwise, search the Library of Congress catalog at https://www.loc.gov.

It has links to large university catalogs.

Start early if you expect to get books on interlibrary loan.

If most sources on your topic are articles, locate a recent one in your library's online databases.

Its database entry will include a list of keywords.

Search for them to find more articles on your topic.

In most cases you can just click on them.

Use the keywords to search the library catalog as well.

Some databases also provide abstracts of journal articles that you can skim for search terms.

3.2.5 Search Guides to Periodical Literature

If you've done research before, you're probably familiar with annual guides e.g. *Readers' Guide to Periodical Literature*, which cites sources e.g. magazines and newspapers.

Most specialized fields also have yearly guides to secondary sources, e.g. *Art Abstracts, Historical Abstracts*, and *Abstracts in Anthropology* (in the bibliography, see items in category 4 in your field).

Most are available online or in other digital forms.

All those resources will direct you to more sources, but none of them can substitute for the kind of in-library search that turns up an unexpectedly useful source.

3.2.6 Prowl the Stacks

Doing research online is faster than on foot, but if you never go into the stacks of your library (assuming you have access), you may miss crucial sources that you'd find only there.

More important, you'll miss the benefits of serendipity - a chance encounter with a source that you find only in person.

If you can get to the stacks, find the self with books on your topic, then scan the titles on that shelf and the ones above, below, and on their side, especially for books with new bindings published by university presses.

Then turn around and skim titles behind you; you never know.

When you spot a promising title, skim its table of contents, then its index for keywords related to your question or its answer.

Then skim its bibliography for titles that look relevant to your project.

You can do all that faster with books on a shelf than you can online.

If the book looks promising, skim its preface or introduction.

If it still looks promising, set it aside for a closer look.

Even if it doesn't seem relevant, record its Library of Congress call number and bibliographic data (author, title, publisher, date of publication and so on; see part 2 of this manual for the details), and in a few words summarize what the book seems to be about.

A month later, you might realize that it's more useful than you thought.

You can check tables of contents for many journals online, but browsing in the journals area of a library can be more productive.

Find the journals that have promising articles.

Skim tables of contents for the prior 10 years.

Most volumes include a yearly summary table of contents.

Then take a quick look at the journals shelved nearby.

Skim their most recent tables of contents.

You will be surprised at how often you find a relevant article that you would have missed had you done your work entirely online.

If you are new to a field, you can get a rough impression of the academic quality of a journal by its look.

If it's on glossy paper with lots of illustrations, even advertisements, it might be more journalistic than scholarly.

Those are not infallible signs of unreliable scholarship, but they are worth considering.

3.2.7 Follow Bibliographic Trails

Most sources will give you trailheads for bibliographic searches.

When you find a book or article that seems useful, skim its bibliography or works cited.

Its index will list the authors cited most often.

Journal articles usually begin with a review of previous research, all cited.

Finally, look for recent PhD dissertations even marginally related to your topic.

Almost every dissertation reviews research in its 1st of 2nd chapter.

By following these bibliographic trails, you can navigate the most difficult research territory, because 1 source leads to another, and another...

3.2.8 Search the Internet

Your practical experience using the internet for everyday research might lead you to believe that it is comprehensive and reliable, but that would be a mistake.

Your library's catalog and databases provide access to information that you cannot get through a search engine.

When doing research online, maintain a healthy skepticism: most of what we retrieve using Google or some other search engine is perfectly reliable, but not everything is.

In contrast to your library's catalog and databases, the internet is essentially unmonitored.

There is no one to vouch for the credibility of materials posted to countless websites.

And finally, keep in mind that companies offering free search engines make their money by acquiring data about you through online behavior and by selling advertising, and that webmasters routinely modify their sites to make them appear higher in search results.

But if you keep these limitations in mind, the internet can be an enormously valuable resource.

In particular, you can do useful preliminary work with a scholarly search engine e.g. Google Scholar or with a tertiary source e.g. Wikipedia (which you should never cite as a reference).

Of course, the internet can also be a primary source if you are researching the internet itself, including social media.

3.2.9 Look for Someone Who Knows Something about Your Topic

You might also ask around or search online to find someone who knows something about the topic and standard reference works on it: advanced students, faculty, even people outside the academic community.

You don't always find someone, but you might get lucky.

3.3 Evaluate Sources for Relevance & Reliability

You will probably find more sources than you can use, so you must evaluate their usefulness by skimming quickly for 2 criteria: relevance and reliability.

3.3.1 Evaluate Sources for Relevance

Once you decide a book might be relevant, do this:

- Skim its index for your keywords, then skim the pages on which those words occur.
- Skim the 1st and last paragraphs in chapters that use a lot of your keywords.
- Skim the book's introduction, especially its last page, where authors and editors often outline their text.
- Skim its last chapter or conclusion, especially the 1st and last several pages.
- Skim prologues, summary chapters, and so on.
- Check the bibliography for titles relevant to your topic.

Be sure that you're looking at a book's most recent edition.

Researchers change their views over time, refining them, even rejecting earlier ones.

If your source is an e-book, still follow these steps, but you can also search the whole text for your keywords.

If you are an advanced researcher, read book reviews of promising sources (see Sect. 4 of the bibliography of resources in your field).

If your source is an journal article, do this:

- Read the abstract, if it has one.
- Skim the introduction and conclusion; if they are not marked off by headings, skim the 1st 6 or 7 paragraphs and the last 4 or 5.
- Skim for section headings, and read the 1st and last paragraphs of those sections.
- Check the bibliography for titles relevant to your topic.

If your source is online, do this:

- If it looks like a printed article, follow the steps for journal article, and also search for your keywords.
- Skim sections labeled "Introduction," "Overview," "Summary," or the like.

  If there are none, look for a link labeled "About the Site" or something similar.
- If the site has a link labeled "Site map" or "Index," check it for your keywords and skim the referenced pages.
- If the site has a search function, type in your keywords.

3.3.2 Evaluate Sources for Reliability

You can't judge a source until you read it, but there are signs of its reliability:

1. **Is the source published by a reputable press?**

   Most books and journals published by university presses, whether in print or in electronic editions, are reliable, especially if you recognize the name of the university.

   You can also trust some commercial presses in some fields, e.g. Norton in literature, Ablex in sciences, or West in the law.

   Be skeptical of a commercial book that makes sensational claims, even if its author has a PhD after his name.

   Be especially careful on contested social issues e.g. gun control or climate change.

   Many books and articles are published by individuals or organizations driven more by ideology than by evidence.

2. **Is the book or article peer reviewed?**

   Most reputable presses and journals ask experts to review a book or article before it is published; this is called *peer review*.

   Essay collections published by university presses are often but not always peer reviewed; someones they are reviewed only the named editor or editors.

   Few commercial magazines use peer review, and fewer still check an author's facts.

   If a book or article hasn't been peer reviewed, use it cautiously.

3. **Is the author a reputable scholar?**

   This is a hard question to answer if you are new to a field.

   Most established scholars are reliable, but use good judgment: even reputable scholars can have axes to grind, especially if their research is supported by a special interest group.

4. **If the source is available only online, is it sponsored by a reputable organization?**

   A website is only as reputable as its sponsor.

   You can usually trust one that is sponsored and maintained by a reputable organization.

   Some sites run by individuals are reliable; most are not.

   Do a web search for the name of the sponsor.

5. **Is the source current?**

   You must use up-to-date sources, but what counts as current depends on the field.

   In computer science, a journal can be out of date in months; in the social sciences, 10 years pushes the limit.

   Publications have a longer shelf life in the humanities: literary or art criticism, e.g., can remain relevant for decades, even centuries.

6. **If the source is a book, dose it have notes and a bibliography?**

   If not, by suspicious, because you have no way to follow up on anything that the source claims.

7. **If the source is a website, does it include bibliographic data?**

   You cannot judge the reliability of a site that does not indicate who sponsors and maintains it, who wrote what's posted there, and when it was posted and last updated.

8. **If the source is a website, does it approach its topic judiciously?**

   Your readers are unlikely to trust a site that engages in heated advocacy, attacks other researchers, makes wild claims, uses abusive language, or makes errors of spelling, punctuation, and grammar.

   The following criteria are especially important for advanced researchers.

9. **If the source is a book, has it been well reviewed?**

   Many fields have indexes to published reviews that tell you how others evaluate a source.

10. **Has the source been frequently cited by others?**

    You can roughly estimate how influential a source is by how often others cite it.

    To determine that, consult a citation index (in the bibliography, see Sect. 4 in your field).

    If you find that a source has been cited repeatedly by other scholars, you can infer that experts in the field regard it as reliable and significant.

    Such sources are said to have an "high impact factor."

Those signs don't guarantee that a source is reliable, but they should give you reasonable confidence in it.

If you can't find reliable sources, acknowledge the limits of the ones you have.

Of course, you may find an exciting research problem when you discover that a source thought to be reliable is not.

3.4 Look beyond the Usual Kinds of References

For a class paper, you'll probably use sources typical in your field.

But if you are doing an advanced project e.g. an MA thesis or PhD dissertation, search beyond them.

If, e.g., you were doing a project on the economic effects of agricultural changes in late 16th-century England, you might read some Elizabethan plays, look at woodcuts of agricultural life, find commentary by religious figures on rural social behavior.

Conversely, if you were working on visual representations of daily life in London, you might work up the economic history of the period and place.

When you look beyond the standard kinds of references relevant to your question, you enrich not only your analysis but also your range of intellectual reference and your ability to synthesize diverse kinds of data, a crucial competence of an inquiring mind.

Don't ignore a work on your topic that is not mentioned in the bibliographies of your most relevant sources - you'll get credit for originality if you turn up good sources others have ignored.

3.5 Record Your Sources Fully, Accurately, & Appropriately

Once you decide a source is worth reading, record all of its bibliographic information.

Your 1st obligation as a researcher is to cite your sources accurately and fully so that your readers can find them.

3.5.1 Determine Your Citation Style

Most fields require a specific citation style.

The 2 most common ones are described in detail in part 2:

- *notes-bibliography style* (or simply *notes style*), used widely in the humanities and in some social sciences (see Chaps. 16–17)
- *author-date style*, used in most social sciences and in the natural sciences (see Chaps. 18–19).

If you are uncertain which style to use, consult your instructor or advisor.

Before you start compiling your list of sources, read the general introduction to citations in Chap. 15 and then, depending on the citation style you are required to use, read the introduction to notes style (Chap. 16) or author-date style (Chap. 18).

3.5.2 Record Complete Bibliographical Information

To save time and avoid errors, record all the citation information you will need when you 1st find a source.

Most of this information appears on the title page of a book or at the head of a journal article.

The specific information you need depends on the type of source, bur for each source, answer the following:

- Who wrote or assembled the source?
  - author(s)
  - editor(s) (if any)
  - translator(s) (if any)
- What data identify the source?
  - title (including subtitle)
  - page numbers (if the source appears in a larger work)
  - volume number
  - issue number
  - edition number
- Who published the source and in what context?
  - publisher's name
  - place of publication
  - name of the journal, collection, or other work in which the source appears
- When was the source published?
  - year of publication
  - season, month, or specific day (and, in some cases, time)
  - for online sources, the date you accessed the material (whether or not you include this date in your citation)
- Where can the source be found?
  - for online sources, a URL or the name of the database
  - for physical items in a 1-of-a-kind collection, the place that houses the collection

For your own use, you might record Library of Congress call numbers.

You won't include them in bibliographic citations, but you may find them helpful if you must consult the source again.

If you access a printed text online, record the URL and any other data for the online version in addition to the full bibliographic data for the original source.

If you scan or photocopy a passage from a book, also scan or photocopy its title page and the bibliographic information on the reverse side.

Then add the library call number if you know it.

At some point, you'll need to format this bibliographic information into your required citation style, so you should record your sources in that style now.

(You can find templates and examples for notes style in Fig. 16.1 and Chap. 17; for author-date style, refer to Fig. 18.1 and Chap. 19.)

For many types of sources, you can copy and paste citation data online if you know where to look.

And there are a number of programs that will help you collect and organize your citations and automatically insert them in your paper in the proper format.

These are useful aids, but they are not perfect, so plan on double-checking your citations not only as you acquire them but also later, after they've been inserted in your paper.

Chapter 4

Engaging Your Sources

Once you have a research problem, use it to guide your research for evidence, models, and arguments to respond to.

But if you don't yet have 1 firmly in hand, you won't know which data, models, or arguments will prove relevant.

So read sources not randomly but deliberately to find a problem.

Look for claims that seem puzzling, inaccurate, or simplistic - anything you can disagree with.

You're more likely to find a research problem when you disagree with a source, but you can also find one in sources you agree with.

Experienced researchers don't read passively; they engage their sources actively, entering into conversation with them.

One you find a source worth a close look, don't read it mechanically, just mining it for data to record.

Note-taking is not clerical work.

When you talk notes on a source thoughtfully, you engage not just its words and ideas but also its implications, consequences, shortcomings, and new possibilities.

4.1 Read Generously to Understand, Then Critically to Engage

If you can, read promising sources twice.

1st, read generously.

Pay attention to what sparks your interest.

Reread passages that puzzle or confuse you.

Don't look for disagreements right away, but read in ways that help the source make sense.

Disagree too soon and you will misunderstand your source or exaggerate a weakness if it presents an argument that challenges yours.

Then, if a source seems important or challenges your own position, read it a 2nd time slowly and more critically.

When you read a passage, think not only about what it says but about how you would respond.

Record those responses in your notes or - if you own the source or a copy of it - in the margins of the source.

Test your understanding by summarizing; if you can't sum up a passage in your mind, you don't understand it well enough to disagree.

You probably won't be able to engage your sources fully until you've done a bit of reading and have developed your own ideas further.

But from the outset, be on alert for ways to engage your sources, actively and creatively.

At some point, the earlier the better, you must look for ways to go *beyond* your sources, too, even when you agree with them.

4.1.1 Look for Creative Agreement

It's a happy moment when a source confirms our views.

But if we just passively agree, we don't develop our own ideas.

So if you believe what a source claims, try to extend that claim: What new cases might it cover?

What new insights can it provide?

Is there confirming evidence the source hasn't considered?

Here are some ways to find a problem through creative agreement.

1. **Offer additional support.** You can offer new evidence to support a source's claim.

Example 4.1.1. *Smith uses anecdotal evidence to show that the Alamo story had mythic status beyond Texas, but editorials in big-city newspapers off better evidence.*

- Source supports _____ with old evidence, but maybe you can offer new evidence.
- Source supports _____ with weak evidence, but maybe you can offer stronger evidence.

2. **Confirm unsupported claims.** You can prove something that a source has only assumed or speculated.

   Example 4.1.2. *Smith recommends visualization to improve sports performance, but a study of the mental activities of athletes shows why that is good advice.*

   - Source speculates _____ might be true, but you can offer evidence to show that it is.
   - Source assumes _____ is true, but maybe you can prove it.

3. **Apply a claim more widely.** You can extend a position to new areas.

   Example 4.1.3. *Smith argues that medical students learn physiological processes better when they are explained with many metaphors rather than with just 1.*

   The same appears to be true for engineers learning physical processes.

   - Source correctly applies _____ to 1 situation, but maybe it can apply to new ones.
   - Source claims that _____ is true in a specific situation, but maybe it's true in general.

4.1.2 Look for Creative Disagreement

If you read actively, you'll inevitably find yourself disagreeing with your sources.

Don't brush those disagreement aside, because they often point to new research problems.

So instead of just noting that you disagree with another writer's views, use that disagreement to encourage your own productive thinking.

Look for these types (the list is not exhaustive, and some overlap).

1. **Contradictions of kind.** A source says something is 1 kind of thing, but maybe it's another kind.

   Example 4.1.4. *Smith says that certain religious groups are considered "cults" because of their strange beliefs, but those beliefs are no different in kind from standard religions.*

   - Source claims that _____ is a kind of _____, but it's not.
   - Source claims that _____ always has _____ as 1 of its features or qualities, but it doesn't.
   - Source claims that _____ is normal/good/significant/useful/moral/interesting, but it's not.

   You can reverse those claims and the ones that follow to state the opposite:

   - Though a source says _____ is *not* a kind of _____, you can show that it is.

2. **Part-whole contradictions.** You can show that a source mistakes how the parts of something are related.

   Example 4.1.5. *Smith has argued that coding is irrelevant to a liberal education, but in fact, it is essential.*

   - Source claims that _____ is a part of _____, but it's not.
   - Source claims that 1 part of _____ relates to another in a certain way, but it doesn't.
   - Source claims that every _____ has _____ as 1 of its parts, but it doesn't.

3. **Developmental or historical contradictions.** You can show that source mistakes the origin and development of a topic.

   Example 4.1.6. *Smith argues that the world population will rise, but it won't.*

   - Source claims that _____ is changing, but it's not.
   - Source claims that _____ originated in _____, but it didn't.
   - Source claims that _____ develops in a certain way, but it doesn't.

4. **External cause-effect relations.** You can show that a source mistakes a causal relationship.

   Example 4.1.7. *Smith claims that legalizing marijuana will increase its use among teenagers, but evidence shows that it doesn't.*

   - Source claims that _____ causes _____, but it doesn't.
   - Source claims that _____ causes _____, but it doesn't; they are both caused by _____.
   - Source claims that _____ is sufficient to cause _____, but it's not.
   - Source claims that _____ causes only _____, but it also causes _____.

5. **Contradictions of perspective.** Most contradictions don't change a conceptual framework, but when you can contradict a "standard" view of things, you urge others to think in a new way.

   Example 4.1.8. *Smith assumes that advertising has only an economic function, but it also serves as a laboratory for new arts forms.*

   - Source discusses _____ from the point of view of _____, but a new context or point of view reveals a new truth. (The new or old context can be social, political, philosophical, historical, economic, ethical, gender specific, etc.)
   - Source analyzes _____ using theory/value system _____, but you can analyze it from a new point of view and see it in a new way.

As the authors said, you probably won't be able to engage sources in these ways until after you've read enough to form some views of your own.

But if you keep these ways of thinking in mind as you begin to read, you'll engage your sources sooner and more productively.

Of course, once you discover that you can productively agree or disagree with a source, you should ask *So what?*

So what if you can show that while Smith claims that easterners did not embrace the story of the Alamo enthusiastically, in fact many did?

4.2 Take Notes Systematically

Once you find and record a source you think you can use, you must read it purposefully and carefully.

Then take notes in a way that will help you not only to remember and use what you have read but also to further your own thinking.

Like the other steps in a research project, note-taking goes better with a plan.

4.2.1 Taking Notes on Paper

Years ago, the standard way to take notes on sources was to create a file of index cards (see Fig. 4.1).

Fig. 4.1. Example of a note card.

At the top left is the author, short title, and page number.

At the top right are keywords that let the researcher sort and re-sort notes into different categories and orders.

The body of the card summarizes the source, records a direct quotation, and includes a comment or thought about further research that is clearly distinguished from the quotation.

At the bottom left is the all number for the source.

A card like this seems old-fashioned, but it provides a template for efficient note-taking:

- Record complete bibliographic information for each source so that you can cite it properly and find it again easily.
- Separate notes on different topics, even if they come from the same source.
- Make sure your notes are accurate, because you need to be able to rely on them later.

  (If you want to quote more than a few lines, copy or save the passage or the whole document.)
- Clearly distinguish
  1. what you quote from a source,
  2. what you paraphrase or summarize from a source, and
  3. your own thoughts.

If you are writing on paper, use headings or brackets or distinct colors to differentiate these 3 different kinds of notes.

You might also create a section specially dedicated to *your own* responses, agreements, disagreements, speculations, and so on (see the italicized section in Fig. 4.1).

That will encourage you to do more than simply record the content of what you read.

4.2.2 Taking Notes Electronically

When you take notes electronically, you have some points:

- You can use a word processor.

  Create a separate file (or at least a separate page) for each source, and be sure to unambiguously distinguish your words from those of your source.

  Though word processors are easy to use, they also limit your ability to index, organize, sort, and search your notes.

  For long or complex projects, you might consider other options.

- You can use a dedicated note-taking application to create and organize your notes.

  Such applications can help you to index, sort, and access your notes.

  But since they sometimes use proprietary formats, they can make it difficult to share your notes or use them with other programs.

- You can use a full-featured citation management program.

  In addition to allowing you to make your own notes, these programs can often pull information directly from online library catalogs and databases, and they can format and update your bibliographies as you write.

  Some will even store full electronic copies of your sources, creating a personal library for your project.

  But like note-taking applications, these programs sometimes use proprietary formats.

In electronic notes, as in notes on paper, you must clearly distinguish your own words and ideas from those of your source.

- Record quotations from your source in a distinctive color or font so that you can recognize them at a glance, and enclose them in large quotation marks in case the file loses its formatting.

- Record paragraphs from your source in a 2nd color or front so that you can't possibly mistake them for your own ideas, and enclose them in curly brackets in case the file loses its formatting.

- Record your own thoughts in a 3rd color or font, and enclose them in square brackets.

  Put longer responses in a separate section so there is no chance you will mistake your own ideas for your source's, or vice versa.

4.2.3 Decide Whether to Summarize, Paraphrase, or Quote

If you can photocopy, scan, or otherwise reproduce your source, or you know you can access it online when you write, you can focus less on preserving its exact words than on your engagement with it.

Summarize the source, which will help you understand it, and note passages you might want to quote or paraphrase when you write.

Note also your response to the source.

Where did you find yourself agreeing or disagreeing with it?

If you can't preserve your source and you don't know whether you will be able to access it later, you have a tougher choice.

It takes too long to transcribe the exact words of every passage you might want to use,so when taking notes, you mus choose as you go whether to quote, paraphrase, or summarize.

Every choice depends on how you plan to use your source:

- Summarize when you need only the point of a passage, section, or even whole article or book.

  Summary is useful for context and for data or claims that are related - but not directly relevant - to your project.

  A summary of a source can never serve as good evidence (see **5.4.2**).

- Paraphrase when the specific words of a passage are less important than its meaning.

  Paraphrase doesn't mean changing just a word or 2.

  You must replace most of the words and phrasing of the original with your own (see **7.9.2**).

  As evidence, a paraphrase is never as good as a direct quotation.

- Record exact quotations for these purposes:

  - The quoted words are evidence that backs up your reasons.
    If, e.g., you claimed that different regions responded to the Battle of the Alamo differently, you would quote exact words from different newspapers.
    You would paraphrase them if you needed only their general sentiments.
  - The words are from an authority you plan to rely on or challenge.
  - The words are strikingly original or so compelling that the quotation can frame the rest of your discussion.
  - The words express a claim that you disagree with, and to be fair you want to state it exactly.

If you don't record important words now, you can't quote them later.

So copy or photocopy more passages than you think you'll need (for more on photocopying, see **4.3.1**).

Never abbreviate a quotation thinking you can accurately reconstruct it later.

You can't.

And if you misquote, you'll damage your credibility.

4.2.4 Guard against Inadvertent Plagiarism

For students and professionals alike, sloppy note-taking has caused grief ranging from ridicule for trivial errors to professional exile for inadvertent plagiarism.

To avoid that risk, commit to heart these 2 iron rules for recording information in notes:

- *Always* unambiguously identify words and ideas from a source so that weeks or months later you cannot possibly mistake them for your own.

  As recommended above, record quotations and paraphrases with quotation marks, as well as in a font or color that unambiguously distinguishes them from your own ideas.

- *Never* paraphrase a source so closely that a reader can match the phrasing and sense of your words with those in your source (see **7.9.2**).

In fact, rather than retyping quotations of more than a few lines, download or photocopy them.

Add to the top of the downloaded or photocopied pages the name of the source and keywords for sorting.

This is important: *never* assume that you can use what you find online without citing its source, even if it's free and publicly available.

You must acknowledge your use of *anything* you did not create yourself.

(For more on plagiarism, see **7.9**.)

4.3 Take Useful Notes

Readers will judge your work not just by the quality of your sources and how accurately you report them but also by how deeply you engage them.

To do that, you must take notes in a way that not only reflects but encourages a growing understanding of your project.

4.3.1 Use Note-Taking to Advance Your Thinking

Many inexperienced researchers think that note-taking is a matter of merely recording data.

Recording or photocopying can help you quote or paraphrase accurately, but if you don't *engage* your sources, you simply accumulate inert data.

To advance your thinking, on any pages you've copied, annotate key sentences and passages by highlighting or labeling them in the margin.

Mark ideas or data that you expect to use in your paper.

Summarize what you have highlighted, or sketch a response to it, or add notes in the margin that help you interpret your highlighting.

The more you write about a source now, the better you will understand it later.

4.3.2 Take Notes Relevant to Your Question & Working Hypothesis

For sources you think especially useful, record not just facts that you think you can use as evidence but also other information that helps you explain those facts and their relationship to your claim.

You can create a template to help with this (see **4.2.1**).

The 1st 3 items directly support or challenge your working hypothesis:

- reasons that support your hypothesis or suggest a new one
- evidence that supports your reasons
- views that undermine or even contradict your hypothesis (see **5.4.3**)

These next items might not support or challenge your hypothesis, but they may help you explain its context or simply make your paper more readable:

- historical background of your question and what authorities have said about it, particularly earlier research (see **6.2.2** an **10.1.1**)
- historical or contemporary context that explains the importance of your question
- important definitions and principles of analysis
- analogies, comparisons, and anecdotes that might not directly support your hypothesis but do explain or illustrate complicated issues or simply make your analysis more interesting
- strikingly original language relevant to your topic.

4.3.3 Get the Context Right

You can't record everything, but you have to record enough to ensure that you accurately capture the source's meaning.

As you use material from sources, record not just what they say but how they use the information.

Here are some ways to guard against misleading your reader.

1. **When you quote, paraphrase, or summarize, be sure to capture the context.**

   When you note an important conclusion, record the author's line of reasoning:

   Example 4.3.1. *Not: Bartolli (p. 123): The war was caused by Z.*

   But Bartolli: The war was caused by X, Y, and Z (p. 123). But the most important cause was Z (p. 123), for 2 reasons: reason 1 (pp. 124–26); reason 2 (p. 126).

   Sometimes you care only about a conclusion, but readers usually want to see how a conclusion emerges from the argument supporting it.

   So when you take notes, record not only conclusions but also the arguments that support them.

2. **When you record a claim, note its role in the original.** Is it a main points?

   A minor point?

   A qualification or concession?

   By noting these distinctions, you avoid this kind of mistake:

   Example 4.3.2. *Original by Jones: "Researchers recognize that lung cancer has a number of causes, including genetic predisposition and exposure to environmental factors e.g. asbestos, radon, and fine particulates. But no one who has studied the data doubts that lung cancer's leading cause is smoking."*

   Misleading report about Jones: Smoking is just 1 cause of lung cancer among many. Jones, e.g., claims that "lung cancer has a number of causes, including genetic predisposition and exposure to environmental factors e.g. asbestos, radon, and fine particulates."

   Jones did not make that point at all.

   He *conceded* a point to set up a point he wanted to make.

   Anyone who deliberately misrepresents an author in this way violates basic standards of truth.

   But you can make such a mistake inadvertently if you note only a source's words and not their role in an argument.

3. **Record the scope & confidence of a claim.** These are not the same:

   Example 4.3.3. *Chemicals in french fries cause cancer.*

   Chemicals in french fries may be a factor in causing cancer.

   Some chemicals in french fries correlate with a higher incidence of cancer.

4. **Don't mistake a summary of another writer's views for those of an author summarizing them.** Some writers do not clearly indicate when they summarize another's argument, so it's easy to quote them as saying what they set out to disprove rather than what they in fact believe.

5. **Note *why* sources agree & disagree.** 2 social scientists might claim that a social problem is caused by personal factors, not by environmental forces, but one might cite evidence from genetic inheritance while the other points to religious beliefs.

   How and why sources agree is as important as the fact that they do.

   In the same way, sources might disagree because they interpret the same evidence differently or take different approaches to the problem.

   It is risky to attach yourself to what any 1 researcher says about an issue.

   It is not "research" when you uncritically summarize another's work.

   If you rely on at least 2 sources, you'll usually find that they do not agree entirely, and that's where your own research can begin.

4.4 Review Your Progress

Regularly review your notes and storyboard to see where you are and where you have to go.

In a storyboard, full pages indicate reasons with support; empty pages indicate work to do.

Consider whether your working hypothesis is still plausible.

Do you have good reasons supporting it?

Good evidence to support those reasons?

Can you add new reasons or evidence?

4.4.1 Search Your Notes for an Answer

The authors have urged you to find a working hypothesis or at least a question to guide your research.

But some writers start with a question so vague that it evaporates as they pursue it.

If that happens to you, search your notes for a generalization that might be a candidate for a working hypothesis, then work backward to find the question it answers.

Look 1st for questions, disagreements, or puzzles in your sources and in your reaction to them (see **2.1.3** and **4.1**).

What surprises you might surprise others.

Try to state that surprise:

Example 4.4.1. *I expected the 1st mythic stories of the Alamo to originate in Texas, but they didn't.*
They originated in . . .

That working hypothesis suggests that the Alamo myth began as a national, not a regional, phenomenon - a modest but promising start.

If you can't find a working hypothesis in your notes, look for a pattern of ideas that might lead you to one.

If you gathered data with a vague question, you probably sorted them under predictable keywords.

For masks, the categories might be their origins (*African, Indian, Japanese,. . .*), uses (*drama, religion, carnival,. . .*), materials (*gold, feather, wood,. . .*), and so on.

E.g.,:

Example 4.4.2. *Egyptians - mummy masks of gold for nobility, wood for others*
Aztecs - masks from gold and jade buried only in the graves of the nobility
New Guinea tribes - masks for the dead from feathers from rare birds

Those facts could support a general statement e.g. *Mask-making cultures use the most valuable materials available to create religious masks, especially for the dead.*

Once you can generate 2 or 3 such statements, try to formulate a still larger generalization that might include them all:

Example 4.4.3. *Many cultures invest great material and human resources in creating masks that represent their deepest values. - generalization*

Egyptians, Aztecs, and Oceanic cultures all created religious masks out of the rarest and most valuable materials.

If you think that some readers might plausibly disagree with that generalization, you might be able to offer it as a claim that corrects their misunderstanding.

4.4.2 Invent Your Question

Now comes a tricky part.

It's like reverse engineering: you've found the answer to a question that you haven't yet asked, so you have to reason backward to invent the question that your new generalization answers.

In this case, it might be *What signs indicate the significance of masks in the societies of those who make and use them?*

As paradoxical as it may seem, experienced researchers often discover their question *after* they answer it, the problem they should have posed after they solve it.

4.4.3 Re-sort Your Notes

If none of that helps, try re-sorting your notes.

When you 1st selected keywords for your notes, you identified general concepts that could organize not just your evidence but your thinking.

If you chose keywords representing those concepts carefully, you can re-sort your notes in different ways to get a new slant on your material.

If your keywords no longer seem relevant, review your notes to create new ones and reshuffle again.

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